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AI Engineer

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Summary

My journey in AI began with building smart solutions that matter — from developing a chatbot for mental health support using LSTM + Attention, to deploying a secure chat interface that prevents data leakage in organizations. I’m driven by the impact of real-time, intelligent systems. Whether it’s visualizing urban taxi trends or clustering customer behavior, I enjoy turning data into action. I thrive at the intersection of AI, software, and real-world problems.

Education

Helwan International Technological University	(HITU)	2022-2026
- Artificial intelligence Department - GPA: 3.78 - Completed coursework: Applied Deep Learning - Completed coursework: Python for AI. - Completed coursework: Python.		

Experience

Freelance AI Developer	Mar 2024 - Present
- Designed and implemented AI-based solutions for various clients in different domains. - Applied techniques such as outlier detection, clustering, and NLP for real-world datasets.	

Projects

IOT Weather Prediction & Smart Control System	Nov-2025 – GitHub-link
- Designed a smart IoT system to monitor temperature, humidity, rain, wind, and light using ESP32 and real sensors. - Trained an ML prediction model to generate “Play/Not Play” decisions based on real-time sensor data. - Created a real-time dashboard to visualize all readings and toggle between Manual and AI control modes.	
Technologies: ESP32, Sensors, PWM RGB Control, Python, ML (Naïve Bayes), php, MySQL Database, Plotly.js, HTML/CSS/JavaScript, IoT Data Pipeline.	
Information Retrieval Engine & Flask web Interface	Nov-2025 – GitHub-link
- Built a complete Information Retrieval pipeline using NLTK preprocessing and Gensim’s TF-IDF model to retrieve, rank, and evaluate over 2,000 movie review documents (POS/NEG). - Deployed a Flask-based search interface displaying top-k results with automated precision, recall, and F1-score evaluation.	
Technologies: Python, NLTK, Gensim, Scikit-learn, Flask, TF-IDF, Cosine Similarity, Information Retrieval, Web Deployment.	
Drug Classification – Evaluation of ML Models & Scaling Techniques	Jul-2025 – GitHub-link
- Built a complete ML pipeline to classify drug types using Decision Tree, SVM, KNN, and Logistic Regression with multiple scaling techniques; evaluated models using accuracy, precision, recall, and F1-score. - Applied feature encoding, scaling (MinMax, Standard, Robust, MaxAbs), and visualized performance with classification reports and decision trees.	
Technologies: Python, Scikit-learn, Pandas, Matplotlib, Seaborn, Classification Models, Label Encoding, Evaluation Metrics	

Mental Health Chatbot

May-2025- [GitHub-link](#)

- Developed an intelligent conversational chatbot as **part of the DEPI Internship Program**, focusing on mental health support and empathetic AI responses.
- Engineered and trained a mental health support chatbot using LSTM with attention, enabling context-aware conversations.
- Deployed via Flask for real-time use, improving accessibility for users in need.

Technologies: Python, Flask, TensorFlow, Keras, Git, GitHub, HTML, CSS, JavaScript, Git LFS.

Sensitive Data Chat Monitor

Apr-2025-[GitHub-link](#)

- Implemented a secure chat interface to prevent data leaks by detecting and blocking sensitive keywords using keyword logic and user intent control.
- Enhanced safety in internal communications through a Flask-Node.js integration.

Technologies: HTML, JavaScript, CSS, Node.js, Express.

Clustering Analysis of Wholesale Customer Data

Apr-2025-[GitHub-link](#)

- Applied clustering techniques (KMeans and DBSCAN) to segment wholesale customers based on annual spending across product categories.
- Performed outlier removal and data standardization to improve model accuracy and robustness.
- Utilized the Elbow Method and Silhouette Score for optimal cluster selection.
- Visualized results using PCA for dimensionality reduction and interpretability.

Technologies: Python, Scikit-learn, Pandas, Matplotlib, Seaborn, PCA, Clustering Metrics.

NYC Taxi Trip Analysis Dashboard

Mar-2025

- Built an interactive analytical dashboard using Plotly Dash to explore NYC taxi trip data, including heatmaps, regression-based fare prediction, and trip duration analysis.
- Enabled real-time data filtering and summary insights.

Technologies: Python, Plotly Dash, Pandas, Mapbox, Scikit-learn, Data Visualization, Linear Regression.

Machine Learning Models Evaluation

Dec-2024 – [GitHub-link](#)

- Python-based tools to analyze and evaluate datasets using machine learning techniques (e.g., classification, regression, and clustering).
- Enhanced decision-making in selecting the best machine learning models for different tasks.

Technologies: python, ML Techniques, Scikit-learn, Pandas, Matplotlib, Seaborn.

House Price Regression: Scaling Method Impact

[Kaggle-link](#)

- Conducted a comparative study on how different feature scaling techniques (Min Max Scaler, Standard Scaler, Robust Scaler, Max Absolute Scaler) affect model performance on house price prediction.
- Evaluated and visualized the performance of Linear Regression and K-Nearest Neighbors (KNN) under each scaling method using R² Score.
- Demonstrated the critical role of preprocessing in non-parametric models like KNN, while showing Linear Regression's stability across scalers.

Technologies: Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Regression, Feature Scaling.

Prosthetic Hand project

4th semester - [Documentary](#)

- Designed a smart prosthetic hand to improve upper-limb functionality for amputees.
- Used IOT and robotics to simulate natural hand movements.

Technologies: ML, IOT technology, Robotics.

Radio Labelling for Radio Stations

Dec-2023 – [GitHub-link](#)

- Solution to interference between communications that arises due to interference from uncontrolled simultaneous transmissions, two channels close to each other can interfere or interact and thus cause damage to communication.

Technologies: python.

K-means implementation

[GitHub-link](#)

- Implementing k-means without using the library, according to the user's choice k to determine the centroid (random, deterministic, which is dividing the data K and choosing the critical point as the centroid), calculating the Euclidean and Manhattan distance.

Technologies: python.

Training & Courses

▪ Generative AI Intern DEPI – CLS	Nov 2025 - Present
▪ AI Intern The British University in Egypt	Feb 2025
▪ AI & ML from Shabab Mubtakirun & Huawei	Jan – Feb 2025
▪ Data science Intern DEPI-AMIT	Oct 2024 – May 2025
▪ Frontend Development course – Udemy	2024
▪ Java course – Udemy	2023
▪ Machine Learning course- Udemy	2022
▪ Python course- Udemy	2022

Skills

- **Technical Skills:**
 - **programming language:** python, Java, JavaScript.
 - Object Oriented Programming (OOP).
 - **Web Development:** HTML, CSS, PHP, Node.js, Django, Flask.
 - **Machine Learning:** Supervised, Unsupervised, Reinforcement Learning.
 - **Deep Learning:** TensorFlow, NLP, LSTM.
 - Data Analysis & Manipulation.
 - **Data Visualization:** Matplotlib, Seaborn, Plotly, Dash.
 - **Big Data Tools:** Apache Spark.
 - Algorithms & Problem Solving.
- **Personal Skills:**
 - problem-solving Abilities.
 - Attention to Detail.
 - Effective Time management.
 - Continuous Learning.